

A Minor Project Synopsis

On

Project ID-BTP2CSE020

Credit Card Fraud Detection

*Submitted in partial fulfillment of the
requirements for the award of the
degree of*

BACHELOR OF TECHNOLOGY



IILM
UNIVERSITY

Session 2025-26
in

Computer Science and Engineering

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March, 2026



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Credit Card Fraud Detection Using Machine Learning

INTRODUCTION

A credit card is a financial tool that allows users to make purchases within a predefined limit and pay later. With the rapid growth of online transactions and digital payments, credit cards have become very common.

However, this growth has also increased the risk of fraud. Credit card fraud happens when someone uses another person's card details without permission. Detecting such fraud is difficult because fraudulent transactions often look similar to normal ones.

PROBLEM STATEMENT

The main problem in credit card fraud detection is that:

- Fraudulent transactions are very rare compared to normal transactions
- This creates an imbalanced dataset
- Machine learning models may fail to correctly identify fraud cases

So, the goal is to build a model that can accurately detect fraudulent transactions.

OBJECTIVES

- To preprocess and clean transaction data
- To apply machine learning algorithms like:
 - Logistic Regression
 - Random Forest
- To handle imbalanced data using SMOTE
- To evaluate models using:
 - Accuracy
 - Precision
 - Recall
 - F1-score

Methodology

1. Data Preprocessing

- Remove missing or irrelevant values
- Normalize data using Standard Scaler

2. Handling Imbalanced Data

- Apply **SMOTE** to increase fraud samples

3. Feature Extraction

- Use transaction details like:
 - Amount

- Time
- Behaviour patterns

4. Model Training

- Train models:
 - Logistic Regression
 - Random Forest

5. Evaluation

- Use confusion matrix and performance metrics

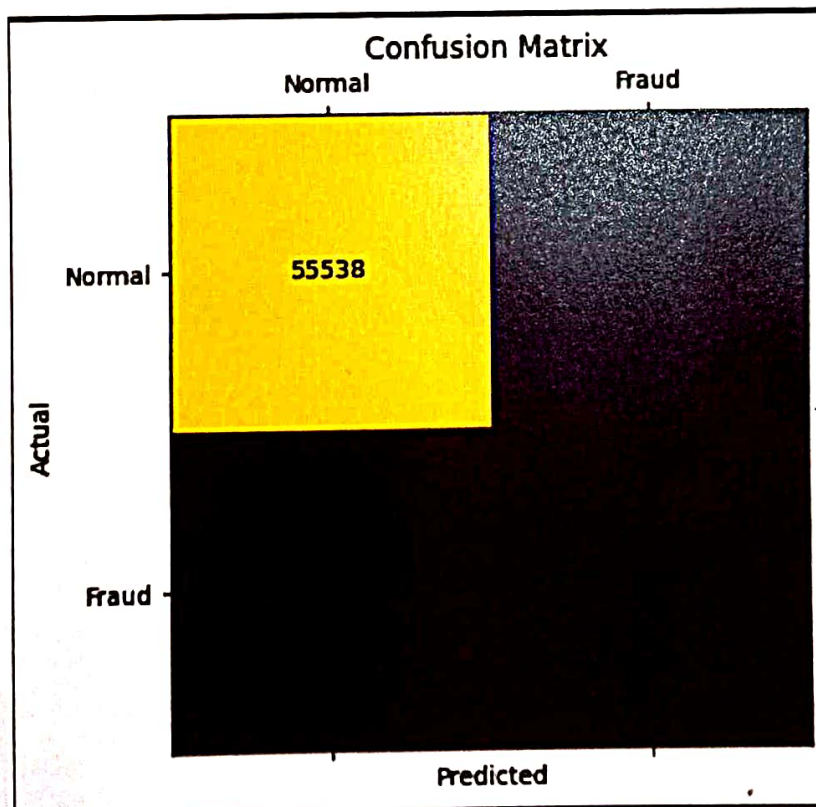
Tools & Technologies

- Python
- NumPy & Pandas
- Matplotlib & Seaborn
- Scikit-learn
- Imbalanced-learn (SMOTE)

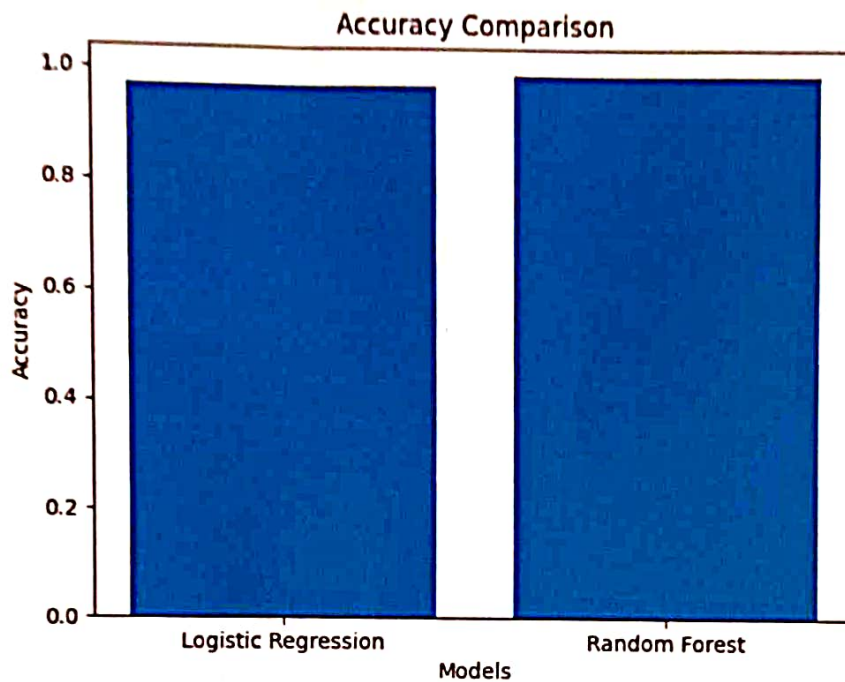
EXPECTED OUTCOMES

Confusion Matrix Example

```
[[55538 1113]
 [ 12 83]]
```



ACCURACY

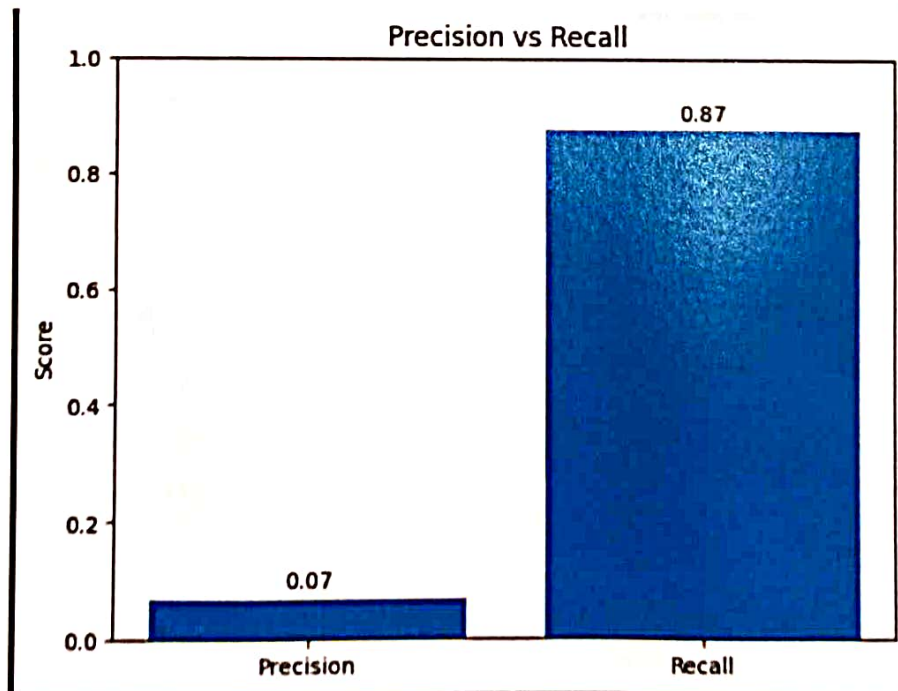


Precision ≈ 0.07 (7%)

Many predicted frauds are actually normal $\rightarrow$ high false alarms

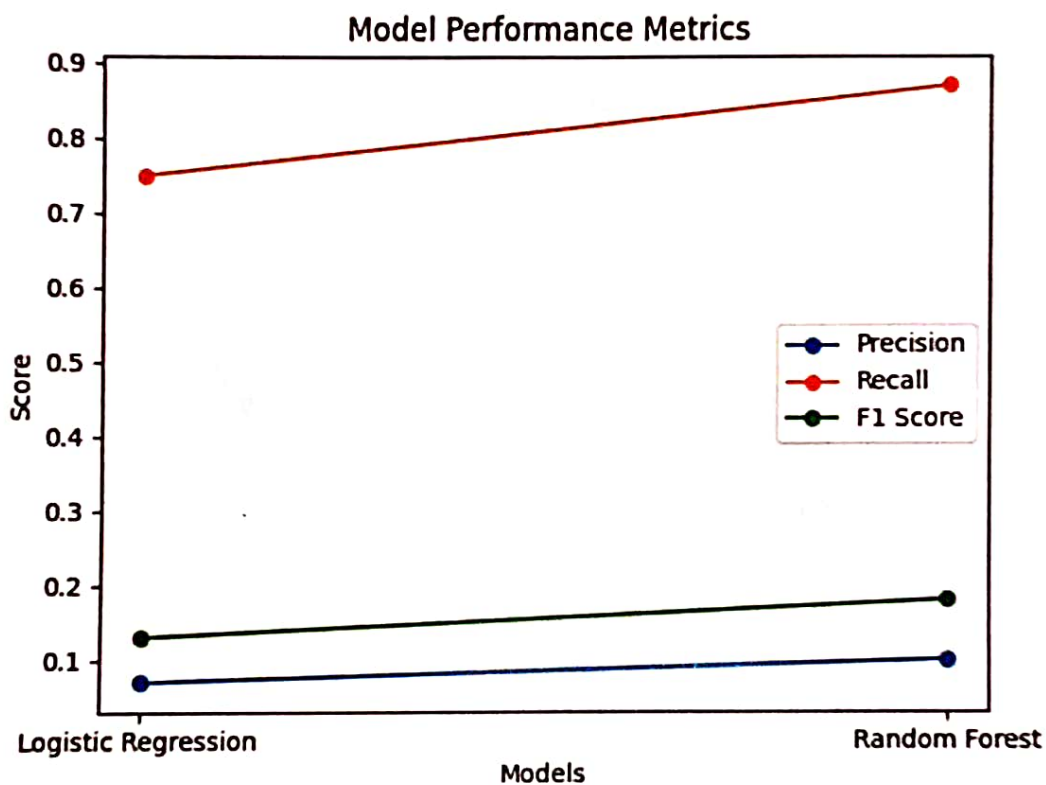
Recall ≈ 0.87 (87%)

Most fraud cases are successfully detected



Model Comparison

- **Logistic Regression**
 - Simple and fast
 - But less effective for complex patterns
- **Random Forest**
 - Handles data better
 - Gives more balanced results
 - Performs better for fraud detection



Final Result: Random Forest performed better.

APPLICATIONS

- Banking fraud detection systems
- Online payment security
- Real-time transaction monitoring
- Fraud alert systems

CONCLUSION AND FUTURE WORK

This project shows how machine learning can help detect credit card fraud.

- Random Forest gave better performance
- Handling imbalanced data is very important

- Accuracy alone is not enough to evaluate the model
- Use advanced models like:
 - XG Boost
 - Neural Networks
- Build live fraud detection systems
- Use deep learning for better accuracy